$$\begin{aligned}
\tau_z|u\rangle &= |u\rangle \\
\tau_z|d\rangle &= -|d\rangle \\
\tau_x|u\rangle &= |d\rangle \\
\tau_x|d\rangle &= |u\rangle \\
\tau_y|u\rangle &= i|d\rangle \\
\tau_y|d\rangle &= -i|u\rangle. \qquad (6.7)
\end{aligned}$$

Now let's consider how the operators should be defined when acting on the tensor product states, |*uu*>, |*ud*>, |*du*>, and |*dd*>. The answer is that when σ acts, it just ignores Bob's half of the state label. There are many possible combinations of operators and states, but I will pick a few at random. You can fill in the others, or look them up in the appendix. Starting with Alice's operators, we find that